



UPI Acquiring Interface Service Technical Specification

Version 1.0

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About this Document

Purpose

To better assist Acquirers outside of Mainland China in implementing UnionPay International (UPI) QR Code (QRC) products, this document specifies requirements for establishing network communication and performing message exchange with UPI system using JSON format.

Audience

This document is intended for Acquirers' IT staff.

Scope

This document covers system security, communication, transaction processing, messages, etc.

Time Expressed

UPI has operation centers in several locations including Shanghai, Beijing, and Hong Kong. The expressions of time in this document, unless particularly indicated, are based on "Beijing time".

Coordinated Universal Time (UTC) is the global standard time by which the world follows. Beijing time is 8 hours ahead of UTC. There is no Daylight Saving Time in China.

Unless otherwise specified, the "day" in this document refers to the calendar day, and the "business day" refers to the working day subject to local regulations of the country/region where the processing Member is located.

References

The terms and conditions in this document have quoted the following publications. Any later versions of these publications will be applied in this document unless otherwise specified.

- *Technical Specifications on Bankcard Interoperability V2.1.*
- [RFC 7159] *The JavaScript Object Notation (JSON) Data Interchange Format*

Support

Please address your questions to the following service team:

Technical Specifications Team

E-mail: ts@unionpayintl.com

Word Conversion

Conversion	Tone	Implication
Shall	Mandatory	Must be followed without exception
Should	Optional but recommended	Required to follow but optional
Must	Mandatory	Required without exception
Must not	Mandatory	Not required
Could	Optional	Hypothetical
Can	Optional	Stating a capability
Cannot	Optional	Stating an incapability
May	Optional	Stating a possibility

Style Conversion

Conversion	Description
------------	-------------

Conversion	Description
Boldface	Note/emphasis
<i>Italic</i>	Publications/book titles

1 Technical Introduction

The UPI Acquiring Interface Service (UAIS) provides robust technical interfaces which allow Acquirers to perform payment with messages in the JSON format. The *UAIS Technical Specification* is designed to specify technical requirements for security, transaction processing and message transmission for the Acquirer system. UPI strongly recommends reading the *Acquirer Implementation Guide for UnionPay QR Code Payment* before this Specification so as to have a better understanding of QRC products.

2 Data Elements and Messages

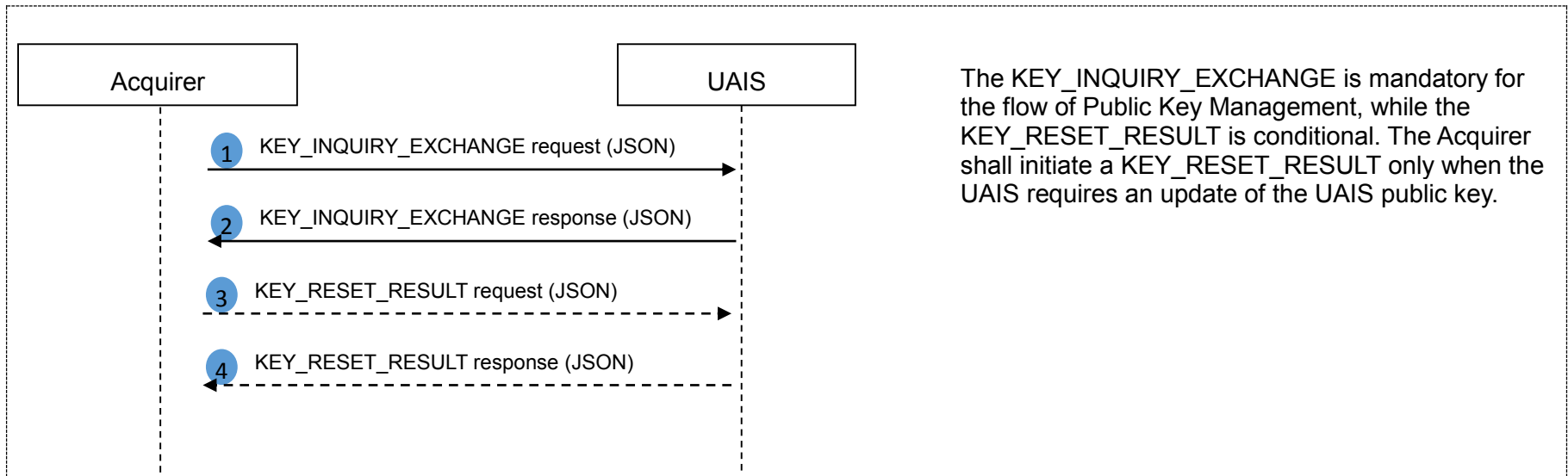
2.1 Message Format

Messages exchanged between the Acquirer system and the UAIS shall be in the JSON data interchange format defined in [RFC 7159].

2.2 Message Flow

The solid line in a processing flow indicates that once the flow begins, the message is mandatory. The dotted line indicates that the message is optional or conditional.

Examples:



2.3 Data Element Attributes

	Description								
Key (K)	The key of the JSON format message.								
Length (L)	The allowed number of characters of the data element value. <u>Examples:</u> - L: 8 indicates the value shall have 8 characters. - L: 15-25 indicates the value shall have a minimum of 15 characters and a maximum of 25 characters.								
Format (F)	The allowed character sets of the data element value. - F: N represents numeric. Data elements in the numeric [N] format shall only contain values represented by digits "0" to "9". The numeric character set includes 10 characters in total. Regardless of the format of the data element, the JSON Data Type shall always be String and encoded in the message by its binary representation as the UTF-8 encoding form. <u>Example:</u> <table border="1"> <tr> <td>Merchant Category Code</td><td>Assuming that the value is "5411", the data element shall be</td></tr> <tr> <td>• K: "mcc"</td><td>"mcc": "5411".</td></tr> <tr> <td>• L: 4</td><td></td></tr> <tr> <td>• F: N</td><td>"mcc": 5411 is not allowed.</td></tr> </table> - F: AN represents alphanumeric. Data elements in the alphanumeric [AN] format shall only contain values represented by "0" to "9", "a"-"z", "A"-"Z" or space. The Alphanumeric character set includes 63 characters in total. - F: ANS represents alphanumeric special. Data elements in the alphanumeric special [ANS] format shall only contain values in the Common Character Set defined in [EMV Book 4]. The Alphanumeric Special character set includes 96	Merchant Category Code	Assuming that the value is "5411", the data element shall be	• K: "mcc"	"mcc": "5411".	• L: 4		• F: N	"mcc": 5411 is not allowed.
Merchant Category Code	Assuming that the value is "5411", the data element shall be								
• K: "mcc"	"mcc": "5411".								
• L: 4									
• F: N	"mcc": 5411 is not allowed.								

	Description
	characters in total, and includes the numeric alphabet and punctuation. • F: S represents string. Data elements in the string [S] format shall contain values in the character set defined in Unicode Standard, the UTF-8 encoding form. • F: array. An array structure is represented by square brackets surrounding zero or more values (or elements). Elements are separated by commas. <u>Example:</u> Cardholder Verification Method "cvm": ["expiryDate", "cvn2", "mobileNo"] • K: "cvm" • L: 1-255 • F: Array

2.4 Data Object

The data object in this document refers to a field that contains multiple data elements.

Examples:

Field Name	Key	Length	Format
Message Response	"msgResponse"		
Response Code	"responseCode"	2	AN
Response Message	"responseMsg"	1-100	S

The data object Message Response contains two data elements: Response Code and Response Message. The data object represented by JSON format shall be:

```
"msgResponse":  
  {"responseCode": "00",  
   "responseMsg": "Approved"  
  }
```

2.5 Presence of Data Element

For the presence of data elements, the following notation is used:

- M: Mandatory – shall always be present.
- C: Conditional – shall be present under certain conditions.
- O: Optional – may be present.

2.6 Data Representation

In this document, characters are enclosed in double quotation marks, for instance "Test@123 银联国际".

A character can be represented by its hexadecimal value. Single quotation marks are used to indicate the hexadecimal value, for instance, '42' to represent the character "B".

2.7 Data Elements Dictionary

2.7.1 Message Information ("msgInfo")

The Message Information shall always be present as the first data object of all request and response messages. The following table defines the data elements of the Message Information.

Field Name		Key	Length	Format	Description	Example															
Message Information		"msgInfo"		Object																	
	Version Number	"versionNo"	5	ANS	<u>Valid Value:</u> "1.0.0"	"1.0.0"															
	Message ID	"msgID"	29	AN	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: <table><tr><td>Generator Indicator</td><td>1 character</td><td>"Q" indicates that the message is generated by the Acquirer system. "P" or "Z" indicates that the message is generated by the UAIS system.</td></tr><tr><td>Acquirer IIN</td><td>8 characters</td><td></td></tr><tr><td>Transmission Year</td><td>4 numeric digits</td><td>The format shall be YYYY</td></tr><tr><td>Transmission Date and Time</td><td>10 numeric digits</td><td>The format shall be MMDDhhmmss</td></tr><tr><td>System Trace Audit number</td><td>6 numeric digits</td><td></td></tr></table>	Generator Indicator	1 character	"Q" indicates that the message is generated by the Acquirer system. "P" or "Z" indicates that the message is generated by the UAIS system.	Acquirer IIN	8 characters		Transmission Year	4 numeric digits	The format shall be YYYY	Transmission Date and Time	10 numeric digits	The format shall be MMDDhhmmss	System Trace Audit number	6 numeric digits		"Q0001034 420171230 235959000 001"
Generator Indicator	1 character	"Q" indicates that the message is generated by the Acquirer system. "P" or "Z" indicates that the message is generated by the UAIS system.																			
Acquirer IIN	8 characters																				
Transmission Year	4 numeric digits	The format shall be YYYY																			
Transmission Date and Time	10 numeric digits	The format shall be MMDDhhmmss																			
System Trace Audit number	6 numeric digits																				

Field Name	Key	Length	Format	Description	Example
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	The value in the response must match the value in the request. The "timeStamp" can be different from the Transmission Year+ Transmission Date and Time in the "msgID"	"20170714235959"
Message Type	"msgType"	1-50	ANS	<u>Valid Value:</u> • PURCHASE • VOID • REFUND • MSG_RESULT_INQUIRY • MPQRC_VERIFICATION • TRX_STATUS_NOTIFICATION • KEY_INQUIRY_EXCHANGE • KEY_RESET_RESULT	"KEY_INQUIRY_EXCHANGE"
Acquirer IIN	"acquirerIIN"	8	AN	The unique value associated to the Acquirer.	"00010344"

2.7.2 Transaction Information ("trxInfo")

The Transaction Information contains different data elements according to the Message Type. The Transaction Information may be present in a request message or a response message. Acquirer can ignore the unexpected Data Elements in the response message.

2.7.3 Message Response ("msgResponse")

The Message Response shall always be present in a response message:

- When it is an acknowledgement response, "responseCode": "00" indicates the request message is well received. The processing result will be sent in another message later.
- When it is a processing result response, "responseCode": "00" indicates the request message is approved.

The following table defines the data elements of the Message Response.

Field Name	Key	Length	Format	Description	Example
Message Response	"msgResponse"		Object		
Response Code	"responseCode"	2	AN	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"
Response Message	"responseMsg"	1-100	S	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the table of Response Code and Message for details.	"Approved"

The valid values of Response Code and Message are shown in the following table. If the message contains a value not shown in the table, the receiver will consider the message rejected, and can display the Response Message as a description.

Table 1: Response Code and Message

Response Code	Response Message
00	Approved
01	Please refer to the card Issuer
03	Invalid Merchant
04	Pending. Transaction result is unknown. Please check later.

Response Code	Response Message
05	Cardholder verification fails.
12	Invalid transaction
13	Invalid amount
14	Invalid card number
15	No such Issuer
21	Card status error
25	Unable to locate the original transaction
30	Message format error
34	Fraud card
40	The transaction is not supported by the Issuer.
41	Lost card
43	Stolen card
51	Insufficient balance
54	Expired card.
57	Transaction not permitted to cardholder
61	Exceeds approval amount limit

Response Code	Response Message
62	Restricted transaction
90	The system is in cut-off.
91	Issuer system error
92	Network error
94	Duplicated transaction
96	UnionPay system error
98	Timeout
A0	Signature verification fails.

2.7.4 Certificate and Signature ("certificateSignature")

The Certificate and Signature shall always be present as the last data object of the request and response messages. The following table defines the data elements of the Certificate and Signature.

Field Name	Key	Length	Format	Description
Certificate and Signature	"certificateSignature"		Object	
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN	The serial number of the UAIS Signature Certificate. The UAIS uses the private key of this certificate to generate a signature, and the Acquirer uses the public key of this certificate to verify the signature. It is present in

Field Name	Key	Length	Format	Description
				the message sent from the UAIS to the Acquirer.
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	The serial number of the Acquirer Signature Certificate. The Acquirer uses the private key of this certificate to generate a signature, and the UAIS uses the public key of this certificate to verify the signature. It is present in the message sent from the Acquirer to the UAIS.
UAIS Encryption Certificate ID	"uaisEncCertID"	1-128	AN	The serial number of the UAIS Encryption Certificate. The Acquirer uses the public key of this certificate to encrypt the sensitive information, and the UAIS uses the private key of this certificate to decrypt it. It is present in the message sent from the Acquirer to the UAIS if encrypted data elements exist in the message.
Acquirer Encryption Certificate ID	"acquirerEncCertID"	1-128	AN	The serial number of the Acquirer Encryption Certificate. The UAIS uses the public key of this certificate to encrypt the sensitive information, and the Acquirer uses the private key of this certificate to decrypt it. It is present in the message sent from the UAIS to the Acquirer if encrypted data elements exist in the message.
Signature	"signature"	1-2048	ANS	Please refer to the section of Signature for details. It is present in all the messages.

3 Security Requirements

3.1 Certificate

The certificate used in the UAIS shall be X.509 standard. RSA key is 2,048 bits in size. The signature certificate shall be different from the encryption certificate.

Examples:

The certificate ID shall be the Serial Number of the certificate. An example is:

```
"uaisSignCertID"="59955a19"
```

The public keys are transmitted in the KEY_INQUIRY_EXCHANGE and KEY_RESET_RESULT. They shall be base64 encoded. An example is:

```
"uaisSignPublicKey"="MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuYZ+Wq0E9fdQbZgt+BOVUd9ea650QITl3xIk  
F3GGn+DuZvjKemDcy7HD3gbFDUjlvooYN4/zLHw1FaFS8VgDnOX4I+lv09Lc9bUz1qcvKN+y0botl6dIXZ13D6NKqtRMEMgUpg  
nRF23r+DjoKaqRATBJg8ZKVRKFtoPeMTuhCpcmEGxIk1MfGMi2SgdohaCXLTFLuh1/SA5mu3DaByE+U2S4UOcEKVwITdKI/xv  
K8D4I19hD+iiWfIYl65xpFFkZhEgs+aFdcj4OtgqyGsJv79JvwwMVTz+grnwUhF3iDKWQuFFFJ5vsftObWqlSeLZ4uj2+XAn68Eg3v  
OKXrGlaQIDAQAB"
```

3.2 Signature

Signature is used to validate the integrity and authenticity of the message. The Acquirer shall follow the steps below to generate a signature or verify the signature:

To sign a message:

1. To-Be-Signed String: Prepare the message to be sent to the UAIS system in the JSON format. Fill the “signature” with “00000000”. Messages shall contain no white space between fields.
2. Use the SHA-256 algorithm to calculate the Hash Value from the To-Be-Signed String, and the Hash Value will be represented as a hexadecimal string in lower case.
3. Use the private key of Acquirer Signature Certificate and RSA (PKCS1_PADDING) algorithm to encrypt the Hash Value.
4. Base64 encode the encrypted Hash Value and put it as the value of “signature”. The Base64 encoding and decoding are defined in [RFC 4648].

To verify a message:

1. To-Be-Signed String: Prepare the message received from the UAIS system in the JSON format. Fill the “signature” with “00000000”. Messages shall contain no white space between fields.
2. Use the SHA-256 algorithm to calculate the Hash Value from the To-Be-Signed String, and the Hash Value will be represented as a hexadecimal string in lower case.
3. Base64 decode the value of “signature” to get the Encrypted Hash Value from the sender.
4. Use the public key of UAIS Signature Certificate and RSA (PKCS1_PADDING) algorithm to decrypt the result from step 3 and check if it is consistent with the result from step 2.

3.3 Encryption

For sensitive information such as Cardholder Verification Information (“cvmInfo”), Primary Account Number (“pan”), etc., the UAIS requires to encrypt its value before transmitting it in the message. The Acquirer shall follow the steps below:

To encrypt a data element:

1. Use the public key of UAIS Encryption Certificate and encrypt a data element with RSA (PKCS1_PADDING) algorithm.
2. Base64 encode the encrypted result and fill it in the value of the field.

To decrypt a data element:

1. Base64 decode the value of the field.
2. Use the private key of Acquirer Encryption Certificate and decrypt the result of step1 with RSA (PKCS1_PADDING) algorithm.

Examples:

Original message

"cvmlInfo":{"expiryDate":"12/22", "cvn2":"123", "mobileNo":"123456789100"}

Data source to be encrypted: {"expiryDate":"12/22", "cvn2":"123", "mobileNo":"123456789100"}

1. Use the UAIS public key of Acquirer Encryption Certificate (key length: 2048 bits) and encrypt the data source with RSA algorithm. (Shown in Hex representation)

79849DE774CC392C238BB8B3C4432744CCFACD880AEA1455B28E15A3A211B0EDC2BA9F7E468F981D846D6F1C411E
BDCF4F193AC0EAF068D28B592D0683AB39B8235F263050F623E47491AA372705173E4424EA2781C450B672DFF71A19
BA8175C6A1E70A593960DBBDFE0F5DE5E2A233C5B3A9626C7062D7BAD27C781E1ACFC9E7C2BF72494FAA337928A4
2F9762B5C1641E7F39FCBE4D54EA79AA09579FB9EF5ECC1057B4C0C27B64E5D14965A1236C706F718A948402553122
B4E71A4AAD406948199C24302BE535E1C8949BF7D8CF0456A2694FCCDF29D3D277BDF5A5B500575260C6C28120D2BA
0CBBA416BDB861623BD2E28EE6C6A4B9FA5F24B1B7D2F9B

2. Base64 encode the encrypted result.

eYSd53TMOSwji7izxEMnRMz6zYgK6hRVso4Vo6IRsO3Cup9+Ro+YHYRtbxxBHR3PTxk6wOrwaNKLWS0Gg6s5uCNfJjBQ9iP
kdJGqNycFFz5EJOongcRQtnLf9xoZuoF1xqHnClk5YNu9/g9d5eKiM8WzqWJscGLXutJ8eB4az8nnwr9ySU+qM3kopC+XYrXB

ZB5/Ofy+TVTqeaoJV5+5717MEFe0wMJ7ZOXRSWWHl2xwb3GKIQCvTEitOcaSq1AaUgZnCQwK+U14ciUm/fYzwRWomlPzN8p09J3vfpbUAV1JgxsKBINK6DLukFr24YWI70uKO5sakufpfJLG30vmw==

3. Fill it in the value of the field.

“cvmlInfo”:

“eYSd53TMOSwji7izxEMnRMz6zYgK6hRVso4Vo6IRsO3Cup9+Ro+YHYRtbxxBHR3PTxk6wOrwaNKLWS0Gg6s5uCNfJjBQ9iPkdJGqNycFFz5EJOongcRQtnLf9xoZuoF1xqHnClk5YNu9/g9d5eKiM8WzqWJscGLXutJ8eB4az8nnwr9ySU+qM3kopC+XYrXBZB5/Ofy+TVTqeaoJV5+5717MEFe0wMJ7ZOXRSWWHl2xwb3GKIQCvTEitOcaSq1AaUgZnCQwK+U14ciUm/fYzwRWomlPzN8p09J3vfpbUAV1JgxsKBINK6DLukFr24YWI70uKO5sakufpfJLG30vmw==”

3.4 Public Key Management

3.4.1 Processing Flow

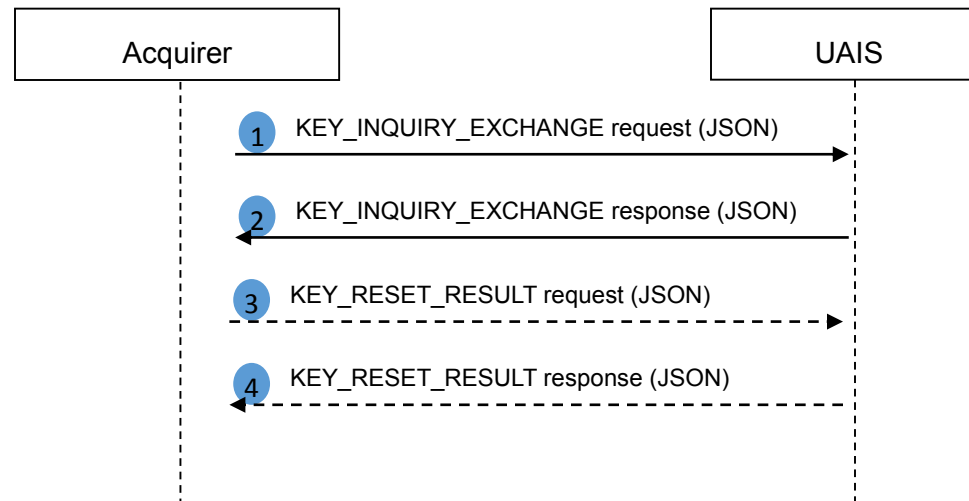


Figure 1: Flow of Public Key Management

1. KEY_INQUIRY_EXCHANGE request

The Acquirer shall submit a KEY_INQUIRY_EXCHANGE request message to the UAIS system at least once per day.

- a. The Acquirer shall submit the UAIS Signature Certificate ID and UAIS Encryption Certificate ID in the Transaction Information so that the UAIS can check if the Certificates need updating.
- b. The Acquirer shall submit the New Acquirer Signature Certificate ID and New Acquirer Signature Public Key in the Transaction Information if they need updating.
- c. The Acquirer shall submit the New Acquirer Encryption Certificate ID and New Acquirer Encryption Public Key in the Transaction Information if they need updating.
- d. The Acquirer shall use the Old Acquirer Signature Certificate ID and Old Acquirer Signature Public Key for the message signature.

2. KEY_INQUIRY_EXCHANGE response

After receiving the KEY_INQUIRY_EXCHANGE request from the Acquirer:

- a. The UAIS will check if the UAIS Signature Certificate ID is the latest ID. If not, the UAIS will return the New UAIS Signature Certificate ID and New UAIS Signature Public Key in the Transaction Information.
- b. The UAIS will check if the UAIS Encryption Certificate ID is the latest ID. If not, the UAIS will return the New UAIS Encryption Certificate ID and New UAIS Encryption Public Key in the Transaction Information.
- c. If the Acquirer submits the New Acquirer Signature Certificate ID or New Acquirer Encryption Certificate ID, the UAIS shall add in the system the New Acquirer Signature Certificate ID and New Acquirer Signature Public Key, or the New Acquirer

Encryption Certificate ID and New Acquirer Encryption Public Key. The UAIS will reject the transaction if it fails to add them in the system.

- d. The UAIS shall use the Old UAIS Signature Certificate ID and Old UAIS Signature Public Key for the response message signature.

After receiving the KEY_INQUIRY_EXCHANGE response:

- a. If the UAIS rejects the KEY_INQUIRY_EXCHANGE request, the Acquirer shall continue to use the Old Acquirer Signature Public Key and Old Acquirer Encryption Public Key for the following payment messages.
- b. If the UAIS approves the KEY_INQUIRY_EXCHANGE request, the Acquirer shall use the New Acquirer Signature Public Key and New Acquirer Encryption Public Key for the following payment messages.

3. KEY_RESET_RESULT request

If the UAIS returns the New UAIS Signature Certificate ID or New UAIS Encryption Certificate ID, the Acquirer shall add in its system the New UAIS Signature Certificate ID and New UAIS Signature Public Key, or the New UAIS Encryption Certificate ID and New UAIS Encryption Public Key. After this is done, the Acquirer shall send the KEY_RESET_RESULT request to the UAIS.

4. KEY_RESET_RESULT response

The UAIS returns KEY_RESET_RESULT response. The UAIS shall use the Old UAIS Signature Certificate ID and Old UAIS Signature Public Key for the response message signature.

Note:

1. During the Public Key Management processing, the UAIS and Acquirer shall both always use the Old Signature Certificate ID and Old Signature Public Key for the “signature” in the request and response messages of KEY_INQUIRY_EXCHANGE and KEY_RESET_RESULT. The New Signature Certificate ID and New Signature Public Key shall be enabled after the Public Key Management processing.

2. Once an encryption public key or a signature public key is updated successfully, both the UAIS and the Acquirer shall reserve the old key and the new key for at least 7 days in their systems. After 7 days, the systems can remove the old key only if the new key is successfully used for one message.

3.4.2 KEY_INQUIRY_EXCHANGE

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	Valid Value: "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time (MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000031"
Time Stamp	"timeStamp "	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	Valid Value: "KEY_INQUIRY_EXCHANGE"	"KEY_INQUIRY_EXCHANGE"
Acquirer IIN	"acquirerIIN "	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Transaction Information	"trxInfo"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN	M	C Present if the UAIS Signature Public Key is updated.	The Acquirer will send the UAIS Signature Certificate ID in the request message. The UAIS will check if it is the latest UAIS signature certificate. If yes, the data element will not be present in the response message. If not, the UAIS will return the new Certificate ID, and the Acquirer gateway shall add the new UAIS Certificate ID and the UAIS Signature Public Key in its system.	
UAIS Signature Public Key	"uaisSignPublicKey"	1-2048	ANS		C Present if the UAIS Signature Public Key is updated.	When the Signature Public Key needs updating, the UAIS will return the latest UAIS Signature Public Key, and the Acquirer shall add it in its system. Otherwise, the data element shall not be present.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	C Present if the Acquirer Signature Public Key is updated.		When the Acquirer Signature Public Key needs updating, the Acquirer shall send the new Acquirer Signature Certificate ID in the request message. Otherwise, the data element shall not be present.	

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Acquirer Signature Public Key	"acquirerSignaturePublicKey"	1-2048	ANS	C Present if the Acquirer Signature Public Key is updated.		When the Acquirer Signature Public Key needs updating, the Acquirer shall send the new Acquirer Signature Public Key in the request message. Otherwise, the data element shall not be present. The UAIS will reject the message if the Acquirer Signature Public Key updating fails for any reason.	
UAIS Encryption Certificate ID	"uaisEncCertificateID"	1-128	AN	M	C Present if the UAIS Encryption Public Key is updated.	The Acquirer will send the UAIS Encryption Certificate ID in the request message. The UAIS will check if it is the new UAIS encryption certificate. If yes, the data element will not be present in the response message. If not, the UAIS will return the latest Certificate ID, and the Acquirer shall add the UAIS Encryption Certificate ID and the UAIS Encryption Public Key in its system.	
UAIS Encryption Public Key	"uaisEncryptionPublicKey"	1-2048	ANS		C Present if the UAIS Encryption Public Key is updated.	When the UAIS Encryption Public Key needs updating, the UAIS will return the latest UAIS Encryption Public Key, and the Acquirer shall add it in its system. Otherwise, the data element will not be present.	
Acquirer Encryption Certificate ID	"acquirerEncryptionCertificateID"	1-128	AN	C Present if the Acquirer Encryption Public Key is updated.		When the Acquirer Encryption Public Key needs updating, the Acquirer shall send the latest Acquirer Encryption Certificate ID in the request message. Otherwise, the data element shall not be present.	

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Acquirer Encryption Public Key	"acquirerEncPublicKey"	1-2048	ANS	C Present if the Acquirer Encryption Public Key is updated.		When the Acquirer Encryption Public Key needs updating, the Acquirer shall send the latest Acquirer Encryption Public Key in the request message. Otherwise, the data element shall not be present. UAIS will reject the message if the Acquirer Encryption Public Key updating fails for any reason.	
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN		M	The serial number of the old UAIS Signature Certificate. The UAIS uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	M		The serial number of the old Acquirer Signature Certificate. The Acquirer gateway uses the private key of this certificate for signature.	

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Signature	"signature"	1-2048	ANS	M	M	The UAIS and Acquirer gateway shall use the Old Signature Public Key for the signature. Please refer to the section of Signature for details.	

3.4.3 KEY_RESET_RESULT

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	Valid Value: "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time (MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000031"
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Type	"msgType"	1-50	ANS	M	M	Valid Value: "KEY_RESET_RESULT"	"KEY_RESET_RESULT"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object				
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN	C Present if the UAIS Signature Public Key is updated.		When it is filled with the new UAIS Signature Certificate ID, it means the Acquirer has updated the new Signature Public Key successfully. When it is filled with the old UAIS Signature Certificate ID, it means the Acquirer has failed to update the new Signature Public Key. It shall be absent when it is not required to be updated.	
UAIS Encryption Certificate ID	"uaisEncCertID"	1-128	AN	C Present if the UAIS Encryption Public Key is updated.		When it is filled with the new UAIS Encryption Certificate ID, it means the Acquirer has updated the new UAIS Encryption Public Key successfully. When it is filled with the old UAIS Encryption Certificate ID, it means the Acquirer has failed to update the new UAIS Encryption Public Key. It shall be absent when it is not required to be updated.	
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"

Field Name	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Response Message	"response Msg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificate Signature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN		M	The serial number of the old UAIS Signature Certificate. The UAIS uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	M		The serial number of the old Acquirer Signature Certificate. The Acquirer gateway uses the private key of this certificate for signature.	
Signature	"signature"	1-2048	ANS	M	M	The UAIS and Acquirer gateway shall use the Old Signature Public Key for the signature. Please refer to the section of Signature for details.	

4 Transaction Processing

This chapter describes the message flows of different transactions.

4.1 Consumer-presented QRC-based Transaction

4.1.1 Consumer-presented QRC-based Purchase

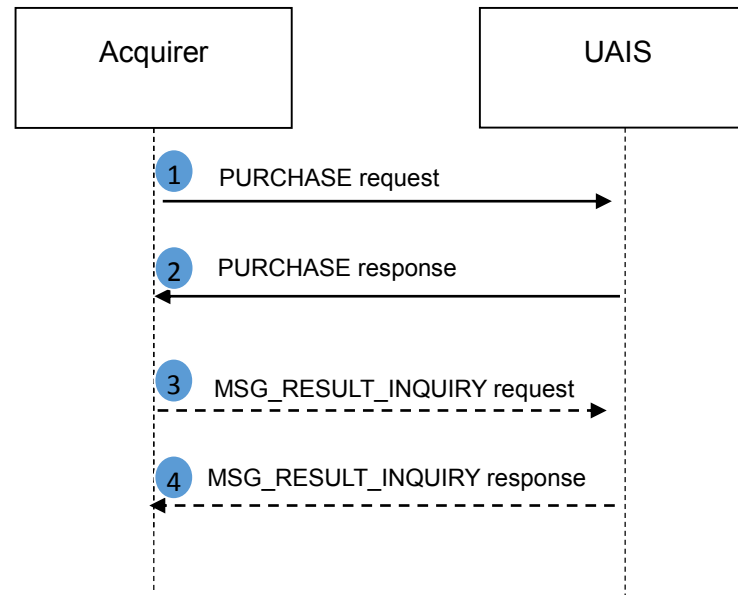


Figure 2: Flow of Consumer-presented QRC-based Purchase

1. The Acquirer submits a PURCHASE request message to the UAIS system.
2. The UAIS system forwards the payment request to the Issuer and brings back the transaction outcome to the Acquirer in the PURCHASE response message.

Exceptional flow:

3. When the Acquirer does not receive the PURCHASE response message within 45 seconds, it can initiate a MSG_RESULT_INQUIRY message, using the Message ID of the PURCHASE to check if the payment is successful.
4. The UAIS system will return the PURCHASE transaction outcome and settlement information to the Acquirer.

4.1.2 Purchase Cancellation

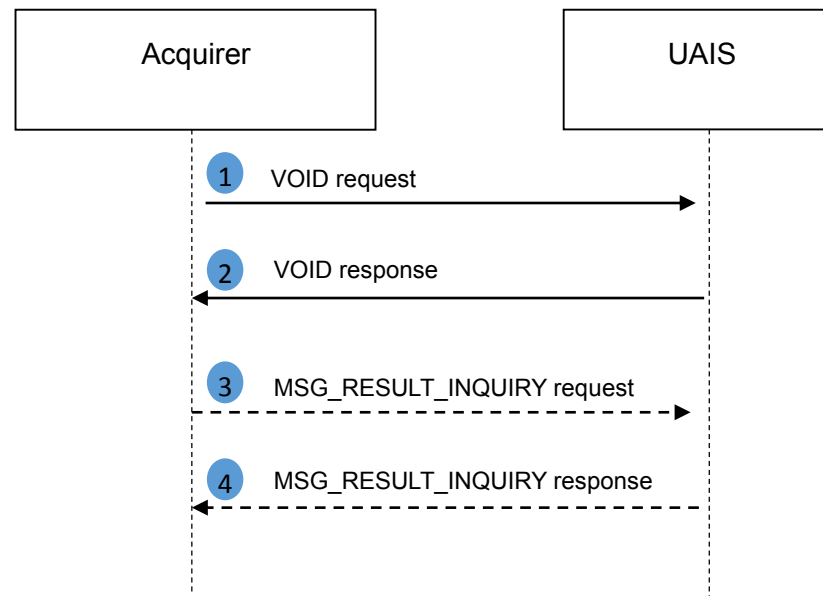


Figure 3: Flow of Purchase Cancellation

1. The Acquirer submits a VOID request message to the UAIS system, using the Message ID of PURCHASE to void the Consumer-presented QRC-based purchase.
2. The UAIS system checks if the original message exists. If not, the UAIS system will reject the transaction. If yes, it will forward the VOID request to the Issuer and bring back the transaction outcome to the Acquirer in the VOID response message.

Exceptional flow:

3. When the Acquirer does not receive the VOID response message within 45 seconds, it can initiate a MSG_RESULT_INQUIRY message, using the Message ID of the VOID to check the transaction status.
4. The UAIS system returns the VOID transaction result to the Acquirer.

4.1.3 Refund

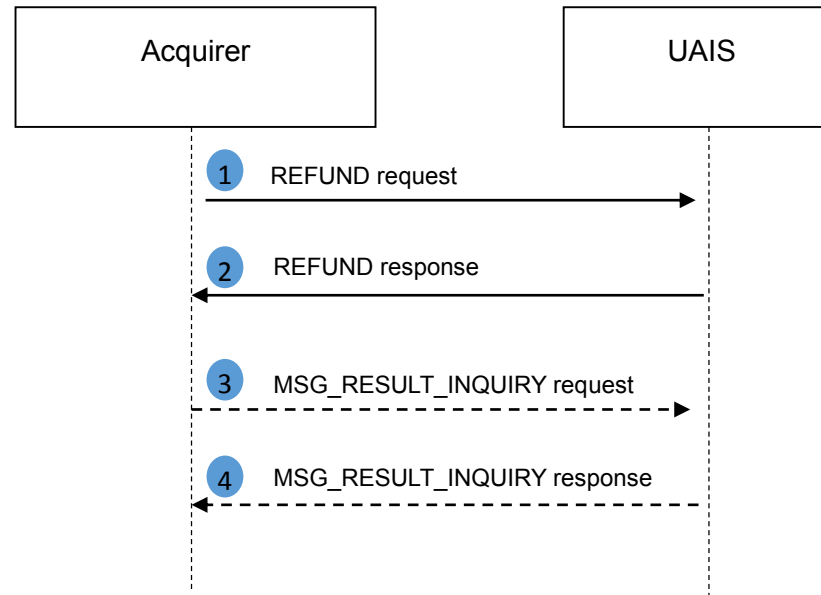


Figure 4: Flow of Refund

1. The Acquirer submits a REFUND request message to the UAIS system.
2. The UAIS system will check if the original payment exists. If not, the UAIS system will reject the transaction. If yes, it will return the transaction outcome to the Acquirer in the REFUND response message.

Exceptional flow:

3. When the Acquirer does not receive the REFUND response message within 45 seconds, it can initiate a MSG_RESULT_INQUIRY message to check the transaction status.

4. The UAIS system will return the transaction result and settlement information to the Acquirer.

4.1.4 Message Result Inquiry

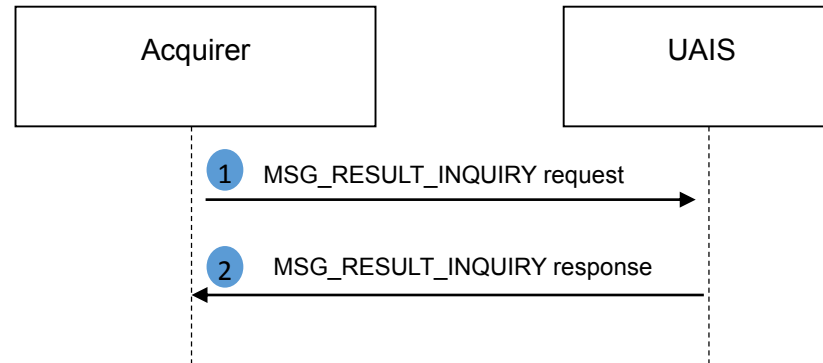


Figure 5: Flow of Message Result Inquiry

1. The Acquirer submits a MSG_RESULT_INQUIRY, using Message ID of the inquired message to check if the message is approved, or rejected.
2. The UAIS system will return the message result to the Acquirer.

Exceptional flow:

The Acquirer can resend the MSG_RESULT_INQUIRY if the response is not received.

4.2 Merchant-presented QRC-based Transaction

4.2.1 Merchant-presented QRC-based Purchase

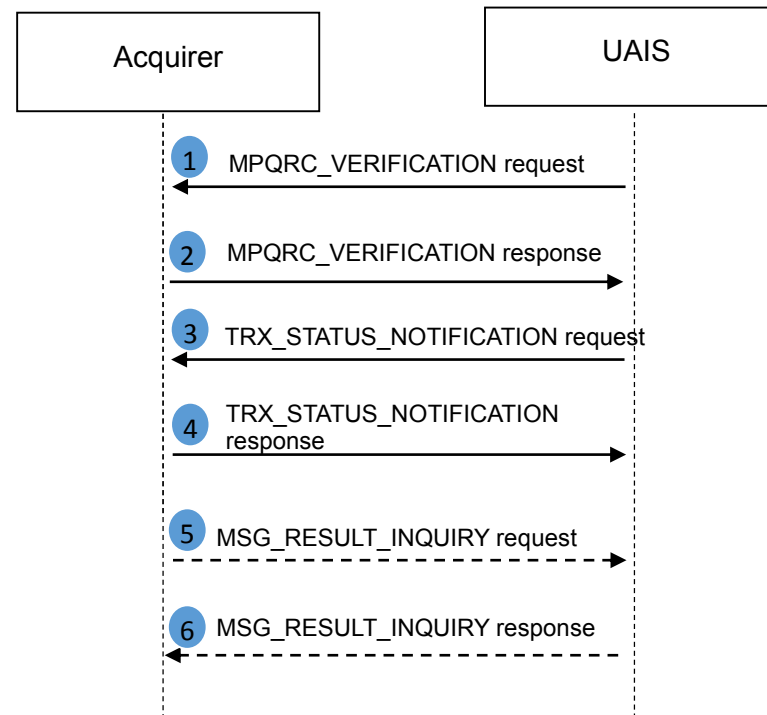


Figure 6: Flow of Purchase

1. When the Consumer opens a mobile application, scans the Merchant-presented QRC and confirms the transaction amount, the mobile application will initiate a purchase request to the UPI network. The UAIS system will then send a MPQRC_VERIFICATION request message to the Acquirer.
2. The Acquirer shall confirm the legitimacy and integrity of the information contained in the QRC, and conduct necessary risk controlling, and return the verification result in the MPQRC_VERIFICATION response message to the UAIS. If the MPQRC_VERIFICATION response message indicates a failed verification, the UPI system will notify the mobile application of the

failed verification, and the transaction processing terminates. If the response from the Acquirer indicates a successful verification of the QRC, the transaction flow proceeds to step 3.

3. The UAIS submits a purchase transaction request to GSCS, and GSCS will forward it to the Issuer for online authorization. After the UAIS receives the purchase response returned by the Issuer, the UAIS will send the transaction result in the TRX_STATUS_NOTIFICATION message to the Acquirer. The Acquirer can match the TRX_STATUS_NOTIFICATION to the MPQRC_VERIFICATION by the QRC voucher number. The original message ID of the TRX_STATUS_NOTIFICATION will also match the message ID of the MPQRC_VERIFICATION
4. The Acquirer returns the TRX_STATUS_NOTIFICATION response to the UAIS system.

Exceptional flow 1:

If the UAIS system does not receive the MPQRC_VERIFICATION response from the Acquirer within 10s, it will reject the Merchant-presented QRC-based purchase request from the application and send the TRX_STATUS_NOTIFICATION message to the Acquirer with the rejected status.

Exceptional flow 2:

If the UAIS system does not receive the TRX_STATUS_NOTIFICATION response from the Acquirer within 10s, it will store and forward the TRX_STATUS_NOTIFICATION request message.

Exceptional flow 3:

5. If the Acquirer does not receive the TRX_STATUS_NOTIFICATION message from the UAIS system within 45s, it can initiate a MSG_RESULT_INQUIRY request to the UAIS, using QRC Voucher Number of the MPQRC_VERIFICATION to check the transaction status.
6. The UAIS returns a MSG_RESULT_INQUIRY response to the Acquirer to indicate whether the payment transaction is approved, rejected or pending.

4.2.2 Purchase Cancellation

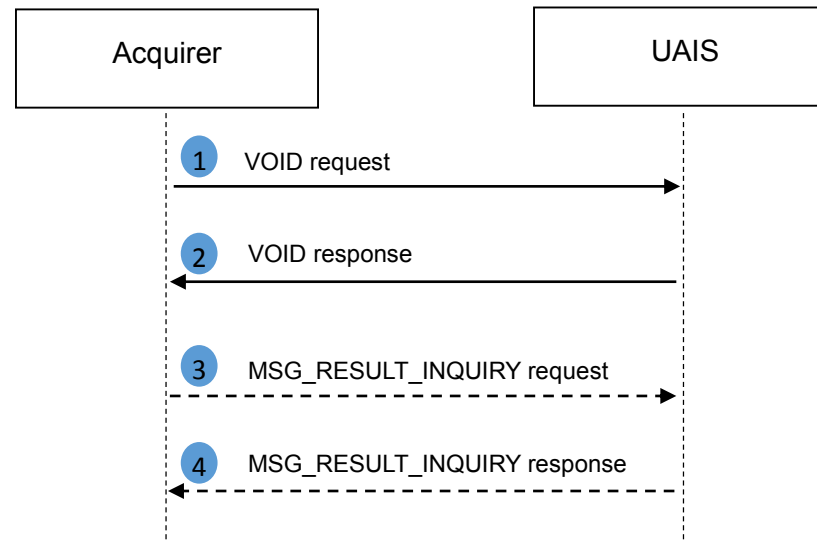


Figure 7: Flow of Purchase Cancellation

1. After the Merchant-presented QRC-based purchase transaction is approved, if the Acquirer intends to cancel the transaction within its settlement day it will submit a VOID request message to the UAIS system, using the QRC voucher number to void the Merchant-presented QRC-based transaction purchase.
2. The UAIS system forwards the purchase cancellation request to the Issuer and returns the transaction result from the Issuer to the Acquirer in the VOID response message.

Exceptional flow:

3. When the Acquirer does not receive the VOID response message within 45 seconds, it can initiate a MSG_RESULT_INQUIRY message, using the Message ID of the VOID to check the result of the VOID message.

4. The UAIS returns a MSG_RESULT_INQUIRY response to the Acquirer to indicate whether the cancellation request message is approved, rejected or pending.

4.2.3 Refund

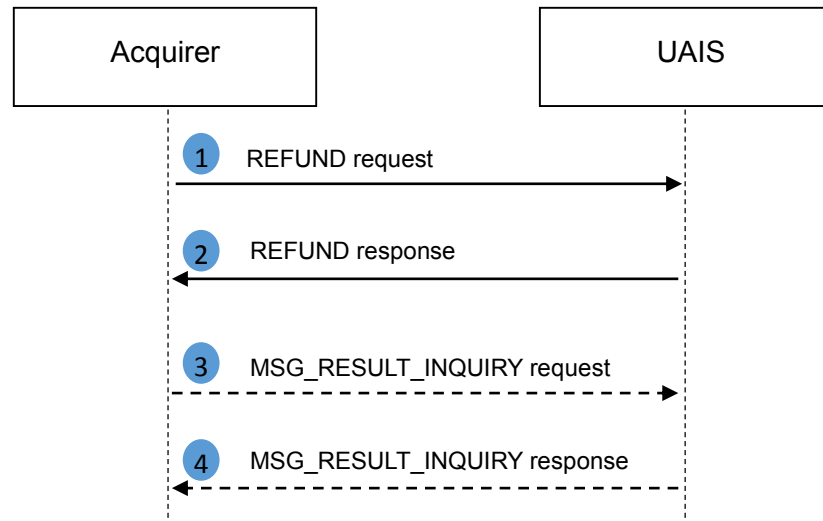


Figure 8: Flow of Refund

1. The Acquirer submits a REFUND request message to the UAIS system in order to initiate a refund transaction.
2. The UAIS system immediately responds to the Acquirer with a REFUND response message. After that, the UAIS system forwards the REFUND request to the Issuer and obtains the REFUND response from the Issuer.

Exceptional flow:

3. If the Acquirer does not receive the REFUND response message within 45 seconds, it can initiate a MSG_RESULT_INQUIRY message, using the Message ID of the REFUND to check the message status.

4. The UAIS returns a MSG_RESULT_INQUIRY response to the Acquirer to indicate whether the refund request is approved, rejected or pending.

4.2.4 Message Result Inquiry

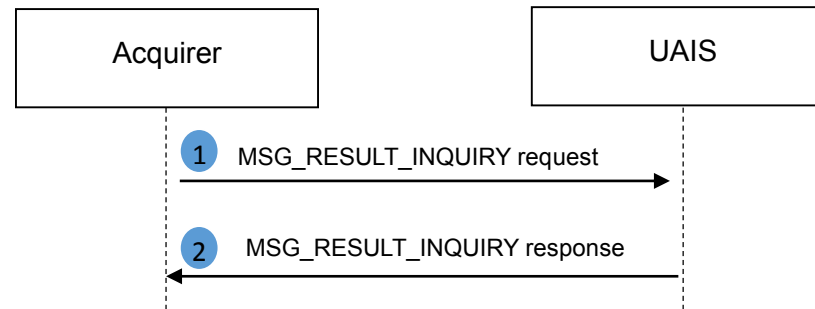


Figure 9: Flow of Message Result Inquiry

1. The Acquirer submits a MSG_RESULT_INQUIRY request message to the UAIS system, using the Message ID of the inquired message to check if the message is approved, or rejected. It can also use the QRC voucher number to check if the Merchant-presented QRC-based payment is approved, or rejected.
2. The UAIS system will return the message result to the Acquirer.

Exceptional flow:

The Acquirer can resend the MSG_RESULT_INQUIRY if the response is not received or the inquired message status is PENDING.

5 Transaction Message

5.1 PURCHASE

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	<u>Valid Value:</u> "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time (MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000000"
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	<u>Valid Value:</u> "PURCHASE"	"PURCHASE"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Card Information Capture Method	"captureMethod"	1-64	ANS	M		<u>Valid Values:</u> "CONSUMER_PRESENTED_QRC" <u>Reserved Values:</u> "NFC" "STORED_CREDENTIALS" "MANUAL" "CONTACT" "MAGSTRIPE" "UNKNOWN"	"CONSUMER_PRESENTED_QRC"
Consumer-presented QRC Payload	"cpqrcPayload"	1-512	ANS	C Present if it is a Consumer-presented QRC-based transaction		It contains the raw payload data.	
Primary Account Number	"pan"	1-2048	AN	M	M	Encrypted with the public key of the Encryption Certificate. The "pan" can be the card number or a token representing the Cardholder account.	Encrypted value of "62111100000000000000"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Transaction Amount	"trxAmt"	1-13	ANS	M	M	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals. The following are examples of valid Transaction Amounts: "98.73", "98" and "98.". The following are NOT valid Transaction Amounts: "98,73" and "3 705". The value must include the transaction fee amount if it exists. The transaction amount in the response message may be unequal to that in the request message due to the discount, which will be used in settlement.	"101.10"
Transaction Currency	"trxCurrency"	3	N	M	M	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the transaction. The currency also applies to the "discount", "costAmount", and "trxFeeAmt" if they exist.	"156"
Transaction Fee Amount	"trxFeeAmt"	1-13	ANS	C Present if the Tip or Convenience Fee exists		The tip or the convenience fee.	"1.10"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Discount Details	"discountDetails"		Array		C Present if the discounts exist.	The details of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	[{"discountAmt": "20", "discountNote": "Uplan discount"}, {"discountAmt": "10", "discountNote": "Instant discount"}]
Discount Amount	"discountAmt"	1-13	ANS		C Present if the discounts exist.	The amount of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	"20"
Discount Note	"discountNote"	1-50	ANS		C Present if the discounts exist.	The note of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	"Uplan discount"
Original Amount	"originalAmount"	1-13	ANS		O	The original amount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	"131.10"
Cost Amount	"costAmount"	1-13	ANS		O	The cost amount after the discount, indicating the amount the Consumer actually pays for the transaction when a promotion is involved. The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from	"101.10"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
						the integer value, and the "." character may be present even if there are no decimals. The cost amount could be displayed to the Consumer in the terminal interface or the printed receipt.	
Coupon Information	"couponInfo"	1-30	ANS	O			
QRC Voucher Number	"qrcVoucherNo"	1-20	ANS		C Populated by UnionPay if the payment is approved.		"1234567890123456789"
Retrieval Reference Number	"retrievalReferenceNumber"	12	AN		C Populated by UnionPay if the payment is approved.		
Merchant ID	"merchantID"	1-15	ANS	M		The unique value associated to the Merchant. The value must uniquely identify any Merchant of the Acquirer.	"ABCDE01"
Merchant Category Code	"mcc"	4	N	M			Convenience store is represented by "5411".
Merchant Country Code	"merchantCountry"	3	N	M			"156"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Merchant Name	"merchantName"	1-25	ANS	M			"UnionPay International"
Merchant City	"merchantCity"	1-15	ANS	M			"Shanghai"
Terminal ID	"terminalID"	1-8	ANS	M		The unique value associated to the terminal.	"abcde01"
Settlement Date	"settlementDate"	4	N (MMDD)		C Populated by UnionPay if the payment is approved.		"1230"
Settlement Currency	"settlementCurrency"	3	N		C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"	A 3-digit numeric value, as defined by [ISO 4217], that indicates the Acquirer's settlement currency.	"840"
Settlement Amount	"settlementAmount"	1-13	ANS		C Populated by	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the	"15.17"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
					UnionPay if the payment is approved and the "settlement Currency" has a different value from "trxCurrency"	amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals.	
Settlement Conversion Rate	"settlementConversionRate"	1-9	ANS		C Populated by UnionPay if the payment is approved and the "settlement Currency" has a different value from "trxCurrency"		"0.15000"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Cardholder Billing Currency	"cardholderBillingCurrency"	3	N		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the card account.	"840"
Cardholder Billing Amount	"cardholderBillingAmt"	1-13	ANS		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals.	"15.17"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Cardholder Billing Conversion Rate	"cardholderBillingConvRate"	1-9	ANS		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"		"0.15000"
Conversion Date	"conversionDate"	4	N (MMDD)		C Populated by UnionPay if either "settlementAmt" or "cardholderBillingAmt" exists		"1230"
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN		M	The serial number of the certificate. The UAIS system uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	M		The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	
Acquirer Encryption Certificate ID	"acquirerEncCertID"	1-128	AN		C Present if encrypted information exists in the request message.	The serial number of the certificate. The UAIS system uses the public key of this certificate for encryption.	
UAIS Encryption Certificate ID	"uaisEncCertID"	1-128	AN	C Present if encrypted information exists in the request message.		The serial number of the certificate. The Acquirer system uses the public key of this certificate for encryption.	

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details.	

5.2 VOID

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	<u>Valid Value:</u> "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time (MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000010"
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	<u>Valid Value:</u> "VOID"	"VOID"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			
Original Message ID	"originalMsgID"	29	AN	M	M	When it is used to cancel the Consumer-presented QRC-based transaction, the value shall be the "msgID" value of the PURCHASE. When it is used to cancel the Merchant-presented QRC-based transaction, the value shall be the "msgID" value of the TRX_STATUS_NOTIFICATION	"Q0001034420171230235959000000"
QRC Voucher Number	"qrcVoucherNo"	1-20	ANS	O		In a VOID, it shall have the same value as in the PURCHASE/TRX_STATUS_NOTIFICATION.	"1234567890123456789"
Transaction Amount	"trxAmt"	1-13	ANS	M		It shall have the same value as in the PURCHASE/TRX_STATUS_NOTIFICATION.	"101.10"
Transaction Currency	"trxCurrency"	3	N	M		It shall have the same value as in the PURCHASE/TRX_STATUS_NOTIFICATION.	"156"
Merchant ID	"merchantID"	1-15	ANS	M		It shall have the same value as in the PURCHASE/TRX_STATUS_NOTIFICATION.	
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN		M	The serial number of the certificate. The UAIS uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	M		The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details.	

Note: Discount information will not be present in VOID messages.

5.3 REFUND

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version	"versionNo"	5	ANS	M	M	<u>Valid Value:</u> "1.0.0"	"1.0.0"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay Acquirer to	Description	Example
Number							
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time(MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000033"
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	<u>Valid Value:</u> "REFUND"	"REFUND"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			
Primary Account Number	"pan"	1-2048	AN	O	M	Encrypted with the public key of the Encryption Certificate. The "pan" can be the card number or a token representing the Cardholder account.	Encrypted value of "62111100000000000000"
Original Message ID	"originalMsgID"	29	AN	M	M	When it is used to refund the Consumer-presented QRC-based transaction, the value shall be the "msgID" value of the PURCHASE. When it is used to refund the	"Q000103442017123023595900000000"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay Acquirer to	Description	Example
						Merchant-presented QRC-based transaction, the value shall be the "msgID" value of the TRX_STATUS_NOTIFICATION	
QRC Voucher Number	"qrcVoucher No"	1-20	ANS	C Present if the original message is a Merchant-presented QRC-based transaction		It is recommended to send QRC voucher number in all the refund message, which shall have the same value as in the PURCHASE or MPQRC_VERIFICATION.	"1234567890123456789"
Transaction Amount	"trxAmt"	1-13	ANS	M	M	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals. The following are examples of valid Transaction Amounts: "98.73", "98" and "98.". The following are NOT valid Transaction Amounts: "98,73" and "3 705". In an approved purchase transaction, the value in the response message indicates the amount approved by the Issuer, which could be less than the value in the request message.	"101.10"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay Acquirer to	Description	Example
Transaction Currency	"trxCurrency"	3	N	M	M	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the transaction. The currency also applies to the "discount", "costAmount", and "trxFeeAmt" if they exist.	"156"
Merchant ID	"merchantID"	1-15	ANS	M			
Merchant Category Code	"mcc"	4	N	M			Convenience store is represented by "5411".
Merchant Country Code	"merchantCountry"	3	N	M			"156"
Merchant Name	"merchantName"	1-25	ANS	M			"UnionPay International"
Merchant City	"merchantCity"	1-15	ANS	M			"Shanghai"
Terminal ID	"terminalID"	1-8	ANS	M			
Retrieval Reference Number	"retrievalReferenceNumber"	12	AN		C Populated by UnionPay if the refund is approved.		
Settlement Date	"settlementDate"	4	N (MMDD)		C Populated by UnionPay if the refund is approved.		"1230"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Settlement Currency	"settlementCurrency"	3	N		C Populated by UnionPay if the refund is approved and the "settlementCurrency" has a different value from "trxCurrency"	A 3-digit numeric value, as defined by [ISO 4217], that indicates the Acquirer's settlement currency.	
Settlement Amount	"settlementAmount"	1-13	ANS		C Populated by UnionPay if the refund is approved and the "settlementCurrency" has a different value from "trxCurrency"	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals.	
Cardholder Billing Currency	"cardholderBillingCurrency"	3	N		C Populated by UnionPay if the refund is approved and the "cardholderBillingCurrency" has a different value from	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the card account.	

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
					"trxCurrency"		
Cardholder Billing Amount	"cardholderBillingAmt"	1-13	ANS		C Populated by UnionPay if the refund is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"		
Settlement Conversion Rate	"settlementConversionRate"	1-9	ANS		C Populated by UnionPay if the refund is approved and the "settlementCurrency" has a different value from "trxCurrency"		"0.15000"
Cardholder Billing Conversion Rate	"cardholderBillingConvRate"	1-9	ANS		C Populated by UnionPay if the refund is approved and the "cardholderBillingCurrency" has a different value		"6.77777"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay Acquirer to	Description	Example
					from "trxCurrency"		
Conversion Date	"conversionDate"	4	N (MMDD)		C Populated by UnionPay if either "settlementAmt" or "cardholderBillingAmt" exists		"1230"
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertificateID"	1-128	AN		M	The serial number of the certificate. The UAIS uses the private key of this certificate for signature.	

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay Acquirer to	Description	Example
Acquirer Signature Certificate ID	"acquirerSig nCertID"	1-128	AN	M		The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	
Acquirer Encryption Certificate ID	"acquirerEnc CertID"	1-128	AN		C Present if encrypted information exists in the request message.	The serial number of the certificate. The UAIS system uses the public key of this certificate for encryption.	
UAIS Encryption Certificate ID	"uaisEncCer tID"	1-128	AN	C Present if encrypted information exists in the request message.		The serial number of the certificate. The Acquirer system uses the public key of this certificate for encryption.	
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details.	

Note: Discount information will not be present in REFUND messages.

5.4 MSG_RESULT_INQUIRY

When the inquired message is financial message such as the Consumer-presented QRC-based purchase or the Merchant-presented QRC-based purchase, the original message response and settlement information will be returned. Otherwise, only the response of the inquired message will be returned.

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	<u>Valid Value:</u> "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that the Acquirer initiates on any day. The value in the response must match the value in the request. Components: "Q"+ Acquirer IIN+ Transmission Year (YYYY) + Transmission Date and Time(MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Q0001034420171230235959000033"
Time Stamp	"timeStamp"	14	N (YYYY MMDD hhmmss)	M	M	The value in response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	<u>Valid Value:</u> "MSG_RESULT_INQUIRY"	"MSG_RESULT_INQUIRY"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			
Original Message ID	"originalMsgID"	29	AN	C One of the "originalMsgID", or "qrcVouche	M	The same value of the "msgID" in the inquired message.	"Q000103441230235959000031"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
				rNo" shall be present.			
QRC Voucher Number	"qrcVoucherNo"	1-20	ANS	C One of the "originalMs glD", or "qrcVoucherNo" shall be present.	C Populated by UnionPay if it exists.	In the request message, the same value in the MPQRC_VERIFICATION message. In the response message, it shall be populated by the UnionPay system.	"1234567890123456789"
Primary Account Number	"pan"	1-2048	AN		M	Encrypted with the public key of the Encryption Certificate. The "pan" can be the card number or a token representing the Cardholder account.	Encrypted value of "62111100000000000000"
Transaction Amount	"trxAmt"	1-13	ANS		C Populated by UnionPay if the payment is approved.	The cost amount of the payment. The transaction amount returned in the MSG_RESULT_INQUIRY message may be unequal to that in the payment request message due to the discount.	"101.10"
Transaction Currency	"trxCurrency"	3	N		C Populated by UnionPay if the payment is approved.	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the transaction. The currency also applies to the "discount", "costAmount", and "trxFeeAmt" if they exist.	"156"
Discount Details	"discountDetails"		Array		C Present if the discounts exist.	The details of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	{{"discountAmt": "20", "discountNote": "Uplan discount"}, {"discountAmt": "10"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
							","discountNote": "Instant discount"]}
	Discount Amount	"discountAmount"	1-13	ANS		C Present if the discounts exist.	The amount of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt. "20"
	Discount Note	"discountNote"	1-50	ANS		C Present if the discounts exist.	The note of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt. "Uplan discount"
	Original Amount	"originalAmount"	1-13	ANS		O	The original amount, which could be displayed to the Consumer in the terminal interface or the printed receipt. "131.10"
	Cost Amount	"costAmount"	1-13	ANS		O	The cost amount after the discount, indicating the amount the Consumer actually pays for the transaction when a promotion is involved. The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals. The cost amount could be displayed to the Consumer in the terminal interface or the printed receipt. "101.10"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Settlement Date	"settlement Date"	4	N (MMD D)		C Populated by UnionPay if the payment is approved.		"1230"
Settlement Currency	"settlement Currency"	3	N		C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"	A 3-digit numeric value, as defined by [ISO 4217], that indicates the Acquirer's settlement currency.	
Settlement Amount	"settlement Amt"	1-13	ANS		C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals.	

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Settlement Conversion Rate	"settlementConvRate"	1-9	ANS		C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"		"6.77777"
Cardholder Billing Currency	"cardholderBillingCurrency"	3	N		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the card account.	
Cardholder Billing Amount	"cardholderBillingAmt"	1-13	ANS		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency"	The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals.	

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
					" has a different value from "trxCurrency"		
Cardholder Billing Conversion Rate	"cardholderBillingConvRate"	1-9	ANS		C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"		"6.77777"
Conversion Date	"conversionDate"	4	N (MMD D)		C Populated by UnionPay if either "settlementAmt" or "cardholderBillingAmt" exists		"1230"
Original Message Response Code	"originalMsgResponseCode"	1-20	ANS		C Present if the "originalMsgID" is presented in the request	Valid Values: • "PENDING" • Response Code, such as "00" When the result of the message inquired is not available, the Message Status shall be "PENDING". Otherwise, it shall	"98"

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
					message	be the "responseCode" of the original message.	
Original Response Message	"originalResponseMessage"	1-100	S		C Present if "originalMsgResponseCode" is presented and the value is not "PENDING"	It shall be the "responseMessage" of the message inquired. Refer to the Response Code and Message for valid values. Refer to the Response Code and Message for details.	"Timeout"
Transaction Response Code	"trxResponseCode"	1-20	ANS		C Present if the "qrcVoucherNo" is presented in the request message	<u>Valid Values:</u> "PENDING" - Response Code, such as "00" It shall be the response code of the Merchant-presented QRC-based transaction result.	"00"
Transaction Response Message	"trxResponseMessage"	1-100	S		C Present if "trxResponseMessage" is presented and the value is not "PENDING"	It shall be the "responseMessage" of the Merchant-presented QRC-based transaction result. Refer to the Response Code and Message for valid values.	"Approved"
Message Response	"msgResponse"				M		

Field	Key	Length	Format	Request Acquirer to UnionPay	Response UnionPay to Acquirer	Description	Example
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed by the Merchant to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN		M	The serial number of the certificate. The UAIS uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN	M		The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	
Acquirer Encryption Certificate ID	"acquirerEncCertID"	1-128	AN		C Present if encrypted information exists in the request message.	The serial number of the certificate. The UAIS system uses the public key of this certificate for encryption.	
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details	

5.5 MPQRC_VERIFICATION

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	Valid Value: "1.0.0"	"1.0.0"
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that UPI initiates on any day. The value in the response must match the value in the request. Components: "Z"+ Acquirer IIN (8 digits)+ Transmission Year (YYYY) + Transmission Date and Time(MMDDhhmmss) + System Trace Audit Number (6 numeric digits)	"Z0001034420171230235959000033"
Time Stamp	"timeStamp"	14	N (YYYYMMDDhhmmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	Valid Value: "MPQRC_VERIFICATION"	"MPQRC_VERIFICATION"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
QRC Voucher Number	"qrcVoucher No"	1-20	ANS	M	M	It is populated by UnionPay system and is used to match a QRC-based transaction in its life cycle. The value must be consistent in the response message, and in the following TRX_STATUS_NOTIFICATION, MSG_RESULT_INQUIRY, VOID, and REFUND messages.	"1234567890123456789"
Merchant-presented QRC Payload	"mpqrcPayload"	1-512	S	M		It contains the raw payload data in the Merchant-presented QRC.	
Card Information Capture Method	"captureMethod"	1-64	ANS	M		<u>Valid Values:</u> • "STORED_CREDENTIALS"	
Primary Account Number	"pan"	1-2048	AN	M	M	Encrypted with the public key of the Encryption Certificate. The "pan" can be the card number or a token representing the Cardholder account.	Encrypted value of "62111100000000000000"
Transaction Amount	"trxAmt"	1-13	ANS	M	M	The value in this field includes tip or convenience fee, if applicable.	"101.1"

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Transaction Currency	"trxCurrency"	3	N	M	M	A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the transaction. The currency also applies to the "discount", "costAmount", and "trxFeeAmt" if they exist.	"156"
Discount Details	"discountDetails"		Array	C Present if the discounts exist.		The details of the discount	[{"discountAmt": "20", "discountNote": "Uplan discount"}, {"discountAmt": "10", "discountNote": "Instant discount"}]
Discount Amount	"discountAmt"	1-13	ANS	C Present if the discounts exist.		The amount of the discount	"20"
Discount Note	"discountNote"	1-50	ANS	C Present if the discounts exist.		The note for the discount	"Uplan discount"

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
QRC Case Use	"qrcUseCase"	2	N		C Present if the Response Code is "00"	<u>Valid Values:</u> "10": purchase transaction "11": purchase transaction, debit card only. Example: purchase finance products. "12": purchase transaction in small businesses. "20": ATM withdrawal. Reserved. "31": P2P QRC-based payment, debit card only. Reserved.	"10"
Additional Data	"additionalData"		Object	C Present if the template "62" exists in the Merchant-presented QRC.			
Bill Number	"billNo"	1-25	ANS	C Present if the ID "01" under template "62" exists in the Merchant-presented QRC.		The invoice number or bill number. It may be provided by the Merchant or entered by the Consumer.	

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
	Mobile Number	"mobileNo"	1-25	ANS	C Present if the ID "02" under template "62" exists in the Merchant-presented QRC.		The Mobile Number to be used for multiple use cases, such as mobile top-up and bill payment. It may be provided by the Merchant or entered by the Consumer. The mobile number may be different from the Consumer mobile number, which will be passed to the Acquirer/Merchant as additional information.
	Store Label	"storeLabel"	1-25	ANS	C Present if the ID "03" under template "63" exists in the Merchant-presented QRC.		A unique value associated to a store. It may be provided by the Merchant or entered by the Consumer.
	Loyalty Number	"loyaltyNumber"	1-25	ANS	C Present if the ID "04" under template "62" exists in the Merchant-presented QRC.		Typically, a loyalty card number. It may be provided by the Merchant or entered by the Consumer.

Field			Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
		Reference Label	"referenceLabel"	1-25	ANS	C Present if the ID "05" under template "62" exists in the Merchant-presented QRC.		Any value defined by the Merchant or Acquirer in order to identify the transaction. It may be provided by the Merchant or entered by the Consumer.	
		Customer Label	"customerLabel"	1-25	ANS	C Present if the ID "06" under template "62" exists in the Merchant-presented QRC.		Any value identifying a specific Consumer. It may be provided by the Merchant or entered by the Consumer.	
		Terminal Label	"terminalLabel"	1-25	ANS	C Present if the ID "07" under template "62" exists in the Merchant-presented QRC.		A unique value associated to a terminal in the store. It may be provided by the Merchant or entered by the Consumer.	

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
	Purpose of Transaction	"trxPurpose"	1-25	S	C Present if the ID "08" under template "62" exists in the Merchant-presented QRC.	Any value defining the purpose of the transaction. It may be provided by the Merchant or entered by the Consumer.	
	Email Address of the Consumer	"consumerEmail"	1-100	S	C Present if the ID "09" under template "62" contains value "E" in the Merchant-presented QRC	The information requested should be provided by the mobile application in the authorization without the need for a prompt pop-up to the Consumer. It will not be verified in the payment procedure, but will be provided to the Acquirer/Merchant as additional information.	
	Address of the Consumer	"consumerAddress"	1-100	S	C Present if the ID "09" under template "62" contains value "A" in the	The information requested should be provided by the mobile application in the authorization without the need for a prompt pop-up to the Consumer. It will not be verified in the payment procedure, but will be provided to the Acquirer/Merchant as additional	

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
				Merchant-presented QRC		information.	
Mobile number of the Consumer	"consumerMobileNo"	1-25	ANS	C Present if the ID "09" under template "62" contains value "E" in the Merchant-presented QRC		The information requested should be provided by the mobile application in the authorization without the need for a prompt pop-up to the Consumer. It will not be verified in the payment procedure, but will be provided to the Acquirer/Merchant as additional information.	
Terminal ID	"terminalID"	1-8	ANS		C Present if the Response Code is "00".		
Merchant Category Code	"mcc"	4	N		O	If the "mcc" is present, the UnionPay system will use the value in this field instead of the MCC in the Merchant-presented QRC Payload for further processing.	

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed on the mobile application to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	
Certificate and Signature	"certificateSignature"		Object	M	M		
UAIS Signature Certificate ID	"uaisSignCertID"	1-128	AN	M		The serial number of the certificate. The UAIS system uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSignCertID"	1-128	AN		M	The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Acquirer Encryption Certificate ID	"acquirerEncCertID"	1-128	AN	C Present if encrypted information exists in the request message.		The serial number of the certificate. The UAIS system uses the public key of this certificate for encryption.	
UAIS Encryption Certificate ID	"uaisEncCertID"	1-128	AN		C Present if encrypted information exists in the response message.	The serial number of the certificate. The Acquirer system uses the public key of this certificate for encryption.	
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details.	

5.6 TRX_STATUS_NOTIFICATION

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Message Information	"msgInfo "		Object	M	M		
Version Number	"versionNo"	5	ANS	M	M	<u>Valid Value:</u> "1.0.0"	"1.0.0"

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Message ID	"msgID"	29	AN	M	M	It is used to match a response to its request. The value must uniquely identify any message that UPI initiates on any day. The value in the response must match the value in the request. Components: "P"+ Acquirer IIN (8 digits)+ Transmission Year+ Transmission Date and Time(MMDDhhmmss) + System Trace Audit Number (6 numeric digits) The Acquirer IIN (8 digits)+ Transmission Year+ Transmission Date and Time(MMDDhhmmss) + System Trace Audit Number (6 numeric digits) is the captured from the "msgID" of MPQRC_VERIFICATION for Merchant-presented QRC-based Purchase	"P000103442017123023595900023"
Time Stamp	"timeStamp"	14	N (YYYYM MDDhh mmss)	M	M	The value in the response must match the value in the request.	"20170714235959"
Message Type	"msgType"	1-50	ANS	M	M	<u>Valid Value:</u> "TRX_STATUS_NOTIFICATION"	"TRX_STATUS_NOTIFICATION"
Acquirer IIN	"acquirerIIN"	8	AN	M	M	The unique value associated to the Acquirer.	"00010344"
Transaction Information	"trxInfo"		Object	M			

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Primary Account Number	"pan"	1-2048	AN	M		Encrypted with the public key of the Encryption Certificate	Encrypted value of "62111100000000000000"
Original Message ID	"originalMsgID"	29	AN	M		the value shall be the "msgID" value of the MPQRC_VERIFICATION	"Q0001034420171230235959000000"
Transaction Amount	"trxAmt"	1-13	ANS	M		The transaction amount. The mobile application can display the transaction amount to the Consumer. The value of the transaction amount shall be the "trxAmt" in the response message of the payment.	"101.10"
Transaction Currency	"trxCurrency"	3	N	M		A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the transaction. The currency also applies to the "discount", "costAmount", and "trxFeeAmt" if they exist.	"156"
QRC Voucher Number	"qrcVoucherNo"	1-20	ANS	M		The same value in the MPQRC_VERIFICATION request message	"1234567890123456789"
Discount Details	"discountDetails"		Array	C Present if the discounts exist.		The details of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	[{"discountAmt": "20", "discountNote": "Uplan discount"}, {"discountAmt": "10", "discountNote": "Instant"}]

Field		Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example	
								discount”}}]	
		Discount Amount	“discountAmt”	1-13	ANS	C Present if the discounts exist.		The amount of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	“20”
		Discount Note	“discountNote”	1-50	ANS	C Present if the discounts exist.		The note of discount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	“Uplan discount”
		Original Amount	"originalAmount"	1-13	ANS	O		The original amount, which could be displayed to the Consumer in the terminal interface or the printed receipt.	“131.10”
		Cost Amount	"costAmount"	1-13	ANS	O		The cost amount after the discount, indicating the amount the Consumer actually pays for the transaction when a promotion is involved. The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present even if there are no decimals. The cost amount could be displayed to the Consumer in the terminal interface or the printed receipt.	“101.10”
		Transaction Response	"trxResponseCode"	1-20	ANS	M		<u>Valid Values:</u> • Response Code, such as “00”	“98”

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Code						It shall be the response code of the Merchant-presented QRC-based transaction result.	
Transaction Response Message	"trxResponseMessage"	1-100	S	M		It shall be the "responseMessage" of the Merchant-presented QRC-based transaction result. Refer to the Response Code and Message for valid values.	"Timeout"
Settlement Date	"settlementDate"	4	N (MMDD)	C Populated by UnionPay if the payment is approved.			"1230"
Settlement Currency	"settlementCurrency"	3	N	C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"		A 3-digit numeric value, as defined by [ISO 4217], that indicates the Acquirer's settlement currency.	"840"
Settlement Amount	"settlementAmt"	1-13	ANS	C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a		The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and the "." character may be present	"15.17"

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
				different value from "trxCurrency"		even if there are no decimals.	
Settlement Conversion Rate	"settlement ConvRate"	1-9	ANS	C Populated by UnionPay if the payment is approved and the "settlementCurrency" has a different value from "trxCurrency"			"0.15000"
Cardholder Billing Currency	"cardholder BillingCurrency"	3	N	C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"		A 3-digit numeric value, as defined by [ISO 4217], that indicates the currency code of the card account.	"840"
Cardholder Billing Amount	"cardholder BillingAmt"	1-13	ANS	C Populated by UnionPay if the payment is approved and the "cardholderBilli		The value shall only include (numeric) digits "0" to "9" and may contain a single "." character as the decimal mark. When the amount includes decimals, the "." character shall be used to separate the decimals from the integer value, and	"15.17"

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
				ngCurrency" has a different value from "trxCurrency"		the "." character may be present even if there are no decimals.	
Cardholder Billing Conversion Rate	"cardholderBillingConvRate"	1-9	ANS	C Populated by UnionPay if the payment is approved and the "cardholderBillingCurrency" has a different value from "trxCurrency"			"0.15000"
Conversion Date	"conversionDate"	4	N (MMDD)	C Populated by UnionPay if either "settlementAmount" or "cardholderBillingAmt" exists			"1230"
Merchant ID	"merchantID"	1-15	ANS	M		The unique value associated to the Merchant. The value must uniquely identify any Merchant of the Acquirer.	"ABCDE01"
Merchant Category Code	"mcc"	4	N	M			Convenience store is represented by "5411".

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
Merchant Country Code	"merchantCountry"	3	N	M			"156"
Merchant Name	"merchantName"	1-25	ANS	M			"UnionPay International"
Merchant City	"merchantCity"	1-15	ANS	M			"Shanghai"
Retrieval Reference Number	"retrievalReferenceNumber"	12	AN	C Populated by UnionPay if the payment is approved.			
Terminal ID	"terminalID"	1-8	ANS	M		The unique value associated to the terminal.	"abcde01"
Message Response	"msgResponse"				M		
Response Code	"responseCode"	2	AN		M	It contains a code that defines the response to a request. Refer to the Response Code and Message for the valid values.	"00"
Response Message	"responseMsg"	1-100	S		M	It contains the transaction result and the rejection reason if the transaction fails. The value of this field can be displayed on the mobile application to notify the Consumer of the payment outcome. Refer to the Response Code and Message for details.	"Approved"
Certificate and Signature	"certificateSignature"		Object	M	M		

Field	Key	Length	Format	Request UnionPay to Acquirer	Response Acquirer to UnionPay	Description	Example
UAIS Signature Certificate ID	"uaisSignC ertID"	1-128	AN	M		The serial number of the certificate. The UAIS uses the private key of this certificate for signature.	
Acquirer Signature Certificate ID	"acquirerSi gnCertID"	1-128	AN		M	The serial number of the certificate. The Acquirer uses the private key of this certificate for signature.	
Acquirer Encryption Certificate ID	"acquirerEn cCertID"	1-128	AN	C Present if encrypted information exists in the request message.		The serial number of the certificate. The UAIS system uses the public key of this certificate for encryption.	
Signature	"signature"	1-2048	ANS	M	M	Refer to the section of Signature for details.	